

AKASH VERMA

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Summary

Final-year B.Tech student skilled in Data Analysis, Machine Learning, and AI-powered application development. Experienced in working with SQL, Python, Pandas, and Power BI for data-driven insights, alongside building end-to-end ML pipelines and LLM-based retrieval systems.

Projects

RAG-Based AI Teaching Assistant

AI / Data Science

- Technologies: Python, Whisper, LangChain, BGE-M3, OpenAI API
- Built a Retrieval-Augmented Generation (RAG) system for answering queries from video lecture content.
- Converted lecture videos into text using OpenAI Whisper and generated semantic chunks.
- Generated dense embeddings using BGE-M3 and implemented cosine similarity-based retrieval.
- Integrated LLM APIs to produce accurate and context-aware responses.

House Price Prediction System

Machine Learning

- Technologies: Python, Scikit-Learn, Pandas, NumPy
- Built an end-to-end machine learning pipeline to predict house prices.
- Performed EDA, handled missing values, feature scaling, and preprocessing.
- Trained and evaluated regression models using RMSE and MAE metrics.
- Persisted trained models using Joblib for deployment.

Casa – AI-Integrated Accommodation Platform

Full Stack + AI

- Technologies: MERN Stack, OpenAI API, Socket.io
- Built a platform for listing, searching, and booking PG accommodations.
- Integrated an AI-powered chatbot for property recommendations and intelligent search.
- Implemented JWT-based authentication and role-based access control.
- Developed a responsive React.js and Tailwind CSS interface with real-time updates.

Technical Skills

Programming Languages: C++, Python, SQL, JavaScript

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap

Backend: Node.js, Express.js, REST APIs, Socket.io

Databases: MongoDB (Mongoose), MySQL

Data Science: NumPy, Pandas, Matplotlib, Seaborn

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Regression, Classification, Clustering

AI / LLM: RAG, Embeddings, BGE-M3, LangChain, Whisper, OpenAI API, Prompt Engineering

Tools: Git, GitHub, VS Code, Google Colab, Joblib, JWT, Postman, Power BI, Excel

Education

Harcourt Butler Technical University

Kanpur, UP

Bachelor of Technology, Electronics Engineering

2022 – Present

CGPA: 6.88/10

St. Peter's School

Rae Bareilly, UP

Class XII (ISC): 90.2%

2022

Class X (ICSE): 81.5%

Key Achievements

- Awarded Gold Medal in District-Level SOF Mathematics Olympiad.
- Won Gold Medal in Table Tennis at Aagaz Sports Fest, HBTU.
- Led the HBTU Table Tennis Team to Quarterfinals at Desportivos Sports Fest, LNMIIT Jaipur.